Guangshen Ma, Ph.D.

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RESEARCH SUMMARY

Physical AI is advancing fast, yet biological tissue remains hard for robots: soft, partly hidden, and deforming when touched. I develop **robotic systems and physical AI algorithms for deformable micro-scale biological tissue**: multimodal perception (OCT, ultrasound, fluorescence, stereo/RGB-D vision), learned deformation models, tissue shape control, and optical beam planning, for medical AI, translational research, and surgical robotics.

ACADEMIC APPOINTMENTS

Postdoctoral Research Fellow, Department of Robotics Feb 2024 – present
University of Michigan, Ann Arbor, MI | Advisor: Mark Draelos, M.D., Ph.D.
Focus: physical AI, medical robotics, multimodal perception (OCT, ultrasound), robot learning, sensor fusion

EDUCATION

Ph.D., Mechanical Engineering Sep 2018 – Dec 2023
Duke University, Durham, NC | Advisor: Patrick J. Codd, M.D.
Dissertation: "3D Tissue Modelling: Laser-based Multi-modal Surface Reconstruction, Crater Shape Prediction and Pathological Mapping in Robotic Surgery"

M.S., Computer Science (concurrent with Ph.D.) Dec 2023
Duke University, Durham, NC

M.S., Mechanical Engineering Sep 2016 – Jun 2018
Duke University, Durham, NC | Advisor: Patrick J. Codd, M.D.
Thesis: "Automatic Brain Surface Registration for the Automated Tumor Resection Device"

B.S., Thermal Energy and Power Engineering Sep 2012 – Jun 2016
Central South University, Changsha, China

HONORS AND AWARDS

Second Prize, Rising Star Poster Competition May 2026
International Symposium on Medical Robotics (ISMR 2026)

Rising Star Recognition, Workshop on the Evolving Landscape of Surgical Robotics May 2025
IEEE International Conference on Robotics and Automation (ICRA 2025)

University Scholarship, Central South University (ranked top 10 of 115 students) 2014–2015

INDUSTRY AND VISITING RESEARCH EXPERIENCE

Research Intern, Camera Autofocus Team, Google May 2022 – Aug 2022
Developed a software prototype for evaluating camera autofocus algorithms.

Visiting Student, Surgical Planning Lab, Brigham and Women's Hospital Jul 2019 – Aug 2019
Boston, MA. Advisors: Jax Luo, Ph.D., and Sarah Frisken, Ph.D.
Project: "Variogram for Checking Problematic Image Registrations" (led to MICCAI 2022 paper, C4)

PUBLICATIONS

* = equal contribution; ‡ = project lead; † = ICRA or IROS paper.

Journal Articles

[J7] K. Koo*, **G. Ma***, X. Wang, and M. Draelos, "A deep learning based robotic ultrasound system for automated probe alignment and precise finger joint localization with underwater light correction," *IEEE Transactions on Medical Robotics and Bionics*, accepted, 2026.

[J6] **G. Ma***, R. Prakash*, B. Schleupner, J. Everitt, A. Mishra, J. Chen, B. Mann, B. Chen, L. Bridgeman, P. Zhong, M. Draelos, W. C. Eward, and P. J. Codd, "TumorMap: A robotic surgical platform for multimodal tumor mapping and laser resection," *npj Digital Surgery* (Nature Portfolio), accepted, 2026. [link]

[J5] **G. Ma**, M. McCloud, Y. Tian, A. Narawane, H. Shi, R. Trout, R. P. McNabb, A. N. Kuo, and M. Draelos, "Robotics and optical coherence tomography: Current works and future perspectives," *Biomedical Optics Express*, vol. 16, no. 2, pp. 578–602, 2025. **(Invited review)**

[J4] X. Peng, K. Koo, **G. Ma**, M. Draelos, and X. Wang, "Photoacoustic imaging combined with robotics: A review of current works," *Biomedical Optics Express*, vol. 16, no. 11, pp. 4730–4745, 2025.

[J3] K. Koo, X. Peng, **G. Ma**, N. Abdulaziz, D. Mills, A. Dentinger, G. Gandikota, M. Draelos, and X. Wang, "Volumetric ultrasound and photoacoustic imaging of inflammatory arthritis in human finger joints via robotic arm powered arc scans," *Biomedical Optics Express*, vol. 16, no. 11, pp. 4746–4766, 2025.

[J2] **G. Ma**, W. Ross, M. Tucker, P.-C. Hsu, D. M. Buckland, and P. J. Codd, "Touch-point detection using thermal video with applications to prevent indirect virus spread," *IEEE Journal of Translational Engineering in Health and Medicine*, vol. 9, pp. 1–11, 2021.

[J1] M. Tucker, **G. Ma**, W. Ross, D. M. Buckland, and P. J. Codd, "Creation of an automated fluorescence guided tumor ablation system," *IEEE Journal of Translational Engineering in Health and Medicine*, vol. 9, pp. 1–9, 2021.

Peer-Reviewed Conference Papers (full length)

[C11] **G. Ma***, T. Qin*, J. Liu, and M. Draelos, "OCT-DeformNet: Optical coherence tomography guided biological tissue shape prediction for robot palpation in microsurgery," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2026, pp. 6168–6175. †

[C10] **G. Ma**, P. J. Codd, and M. Draelos, "Robotic relay of free-space optical beams for medical applications," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2026, pp. 2060–2067. †

[C9] G. Zhou, Y. Wang, Y. Ling, **G. Ma**, R. P. McNabb, A. N. Kuo, N. Valikodath, and M. Draelos, "Imaging eyes in motion: Dynamic compensation with robotic optical coherence tomography," in *Proc. IEEE International Symposium on Medical Robotics (ISMR)*, 2026, pp. 1–7.

[C8] J. Liu, **G. Ma**, G. Zhou, H. Pan, C. Lam, C. Jin, N. Valikodath, and M. Draelos, "Dual-arm teleoperated robotic microsurgery system with real-time volumetric OCT image feedback," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025, pp. 21423–21430. †

[C7] **G. Ma** and M. Draelos, "Geometry-aware volumetric data stitching using local surface mapping and robot optical coherence tomography," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 3788–3794. †

[C6] **G. Ma**, R. Prakash, B. Mann, W. Ross, and P. J. Codd, "3D laser-and-tissue agnostic data-driven method for robotic laser surgical planning," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023, pp. 8446–8453. †

[C5] **G. Ma**, W. Ross, and P. J. Codd, "N-mirror robot system for laser surgery: A simulation study," in *Proc. IEEE International Symposium on Medical Robotics (ISMR)*, 2023, pp. 1–7.

[C4] J. Luo, **G. Ma**, et al., W. M. Wells III, and S. Frisken, "On the dataset quality control for image registration evaluation," in *Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2022, pp. 36– 45.

[C3] **G. Ma**, W. Ross, and P. J. Codd, "StereoCNC: A stereovision-guided robotic laser system," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 540–547. †

[C2] **G. Ma***, S. R. Oca*, Y. Zhu, P. J. Codd, and D. M. Buckland, "A novel robotic system for ultrasound-guided peripheral vascular localization," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 12321–12327. †

[C1] **G. Ma***, W. A. Ross*, I. Hill, N. Narasimhan, and P. J. Codd, "A novel laser scalpel system for computer-assisted laser surgery," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 386–392. †

Manuscripts Under Review and Preprints

[R5] K. Koo*, **G. Ma***‡, X. Wang, and M. Draelos, "Tracker-free robotic ultrasound calibration with a spherical-marker phantom and threshold-free center localization," submitted to ICRA 2027; arXiv:2609.22483. [link]

[R4] C. Zhang*, **G. Ma***‡, and M. Draelos, "Robotic optical coherence tomography-guided geometric modeling of soft tissue: 3D representations for data-driven closed-loop shape control," under review; arXiv:2610.32961.

Additional co-authored journal manuscripts under review are not listed.

Conference Proceedings and Abstracts

[A6] G. Zhou, Y. Wang, Y. Ling, **G. Ma**, R. P. McNabb, A. N. Kuo, N. Valikodath, and M. Draelos, "Retinal imaging during large-scale head and body movement with motion-predictive OCT," *Investigative Ophthalmology & Visual Science*, vol. 67, no. 7, p. 4460, 2026.

[A5] **G. Ma**, K. Koo, X. Peng, M. Draelos, and X. Wang, "High-speed robotic ultrasound scanning for finger joint examination," in *Proc. SPIE*, Photons Plus Ultrasound: Imaging and Sensing, 2026, PC138510P.

[A4] K. Koo, X. Peng, **G. Ma**, M. Draelos, and X. Wang, "Enhanced diagnostic imaging for rheumatoid arthritis in the finger via robotic arc ultrasound scanning and 3D reconstruction," in *Proc. SPIE*, Photons Plus Ultrasound: Imaging and Sensing, 2026, PC138510X. **(Best Paper Award Finalist)**

[A3] **G. Ma**, W. Ross, M. Tucker, and P. J. Codd, "Characterization of photoablation versus incidence angle in soft tissue laser surgery: An experimental phantom study," in *Proc. SPIE 11238*, Optical Interactions with Tissue and Cells XXXI, 2020, pp. 69–80.

[A2] W. Ross, M. Tucker, **G. Ma**, and P. J. Codd, "Model for and analysis of intraoperative brain tumor boundary detection based on known spectral signatures of glioblastoma," in *Proc. SPIE 11229*, Advanced Biomedical and Clinical Diagnostic and Surgical Guidance Systems XVIII, 2020, pp. 60–66.

[A1] M. B. Tucker, S. Joseph, W. Ross, **G. Ma**, P. Chongsathidkiet, P. Fecci, and P. J. Codd, "Creation of a non-contact, automated brain tumor detection device for use in brain tumor resection," in *Proc. SPIE 11225*, Clinical and Translational Neurophotonics, 2020, pp. 24–29.

PATENT APPLICATIONS

[P3] S. Oca, **G. Ma**, D. Buckland, and P. J. Codd, "Systems and methods for autonomous medical intervention," U.S. Patent Application 18/166,111, Duke University, 2023.

[P2] **G. Ma**, W. Ross, and P. J. Codd, "Systems and methods for 3D data-driven laser orientation planning," U.S. Patent Application 18/084,777, Duke University, 2023.

[P1] J. Patel, W. Ross, M. Tucker, **G. Ma**, and P. J. Codd, "Laser system and methods for cutaneous treatments and surgery," U.S. Patent Application 17/928,505, Duke University, 2023.

TEACHING EXPERIENCE

Teaching Assistant, Duke University: Control of Dynamic Systems; Structure and Properties of Solids

PROFESSIONAL SERVICE

Reviewer: IEEE Robotics and Automation Letters (RA-L); IEEE ICRA; IEEE/RSJ IROS; ICLR